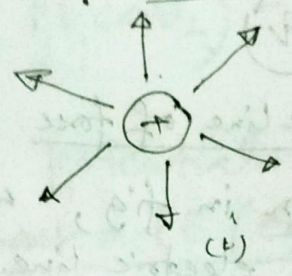
① Electric line of force

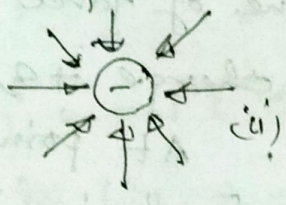
Electric lines of forces are imaginary / hypothetical lines in electric field which give the information about net electric field in that region.

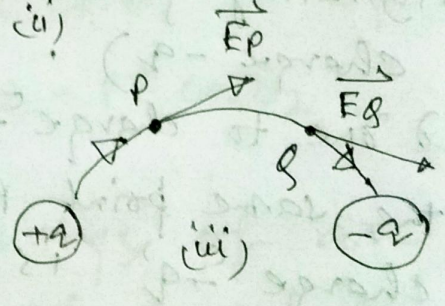
- It is a path on which a positive test charge will tend to move if free to do so.
- It begins from a +ve charge

and

end on a -ve charge





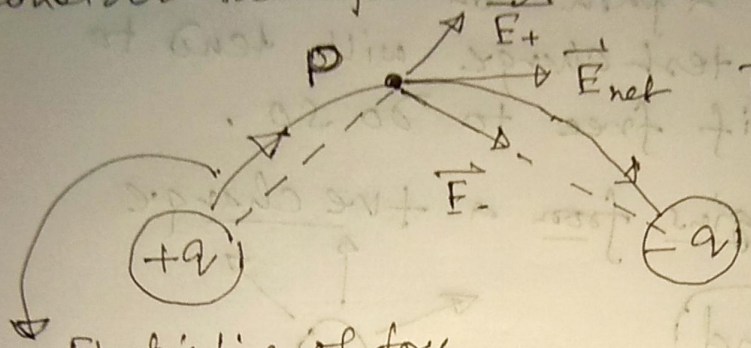


- Tangent to electric line of force at any point gives the direction of net electric field at that point.

In fig (iii) $\vec{E_P}$ and $\vec{E_Q}$ are net electric fields at point P and Q, (of the electric field due to +q & −q). And they are along the tangents to electric line of force at point P & Q.

(2) Concept of Electric line of force or field lines :→

In an electric field, the electric lines of forces gives the net electric field of the region. To understand this concept consider the following example :



$\vec{E}_{net} = \vec{E}_+ + \vec{E}_-$

Electric line of force

As shown in fig, we consider a point P on an electric line of force, (originating from charge +q & terminating on charge -q). At point P, electric field due to charge '+q' is $\vec{E}_+$ and at the same point P, electric field due to charge '-q' is $\vec{E}_-$ (as shown in fig).

The resultant electric field at point 'P' is $\vec{E}_{net}$ $(= \vec{E}_+ + \vec{E}_-)$ is in the direction along the tangent at 'P' to the electric line of force at pt 'P'.

Thus we can say that at every point in electric field, tangent to electric line of force gives the direction of net electric field at that point.